

RCXLV.LLC — Context Dump / Technical + Conceptual Cheat Sheet

This document is intended to provide full situational context for an external coding assistant (e.g. Claude Code). It describes the purpose, structure, philosophy, current implementation state, known bugs, and project-specific details of the rcxlv.llc website and its showcased works.

Audience: coding assistants, collaborators, or future self. Tone: factual, explicit, non-polished.

1. High-Level Purpose of the Website

rcxlv.llc is a portfolio website primarily built for graduate school applications, secondarily as a long-term home for interactive, technical, and artistic work. Core goals:

- Present technically complex creative systems in a calm, legible way.
- Allow interactive works (p5.js, Processing ports, Babylon.js, audio systems) to run live in-browser.
- Avoid traditional “scroll portfolio” patterns.
- Emphasize systems-thinking, experimentation, and research-minded work.
- Be usable by admissions committees without explanation.

2. Conceptual Structure of the Site

The site is structured like a file explorer rather than a traditional website.

Left side:

- A persistent tree navigation.
- Nested folders and numbered sections (01–05).
- Items can be: - Pages (internal HTML microapps) - External links (YouTube, GitHub, Twitch)

Right side:

- A “viewer” pane that hosts preloaded iframes.
- Switching pages is instantaneous (no reload).
- Interactive sketches continue running while hidden.

3. Architectural Decisions (Current MVP)

- Single-page shell with: - Sidebar - Viewer
- Navigation is state-driven via JavaScript.
- Each page is its own self-contained microapp (HTML + JS + CSS).
- Pages are preloaded into hidden iframes on boot.
- Routing is handled client-side via `history.pushState`.
- Direct URLs exist (`/home`, `/cv`, `/lifebrush`), but refresh behavior is currently broken on static hosting.

4. Known Router Limitation (Explicitly Unresolved)

IMPORTANT: Page refresh on deep routes (e.g. `/cv`, `/readme-02`) causes file-not-found errors when statically hosted. This is known, acknowledged, and intentionally unresolved. Development proceeds as if this is not a problem. Solution likely involves server-side rewrites or hash routing, but is deferred.

5. Sidebar System

- Sidebar is generated from `manifest.json`.
- No hardcoded HTML tree.
- Supports: - Folder nodes (emoji folders, click-to-toggle, no caret) - Section nodes (numbered, caret toggle) - Page nodes - External link nodes
- Sidebar can be programmatically controlled by pages (planned).

6. Viewer System

- Viewer holds multiple iframes, one per page.
- Iframes are created eagerly for important pages.
- Pages are shown/hidden via CSS (display: none/block).
- No iframe reload on navigation.
- Interactive state persists across navigation.

7. Lifebrush (rcx04) — Interactive p5.js System

Lifebrush is a persistent, interactive Game-of-Life-based drawing system. Key traits:

- Canvas persists across navigation.
- Grid resolution randomized per interaction.
- Mouse/touch drawing accumulates cells.
- Layers stack over time.
- Clicking resets simulation with memory saved.
- Right click / space randomizes brush color.
- Mobile-aware input handling.

Known quirk:

- Canvas does not receive mouse hover until clicked in some browsers.
- Works in VS Code live preview but not standard browser.
- Focus / iframe interaction issue suspected.
- Considered non-blocking.

8. Project Index Overview

rcx01 — Explorative Loop Sampler • Large-scale audio system. • Keyboard-driven performance interface. • Playtest studies conducted. • Alpha build exists. • Documentation incomplete. • Demos partially outdated.

rcx02 — Installation / Capstone • Multiple demos exist. • No README yet. • No public code. • VOD exists. • Potential livestream idea.

rcx03 — Smaller experimental system • Dinky demos. • Needs cleanup and README.

rcx04 — Lifebrush • Fully functional interactive demo. • No documentation yet.

rcx05 — Processing / TouchDesigner hybrid • One demo posted. • Second demo missing. • No README. • Code not packaged.

rcx06 — Experimental system • Cool but incomplete. • No demos posted. • No documentation.

9. LoopMaker / Research Work

LoopMaker is a research-driven system for loop detection and segmentation in music. Functional features include:

- Source separation (Demucs)
- Beat and downbeat tracking
- Self-similarity matrices
- Novelty detection
- Boundary refinement
- Loop candidate ranking
- Export of audio, metadata, plots, images
- Manual adjustment tools

Status:

- Technically functional.
- Research paper incomplete.
- Interactive paper version planned for website.

10. External Links System

External links (YouTube, GitHub, Twitch) are treated as first-class tree items.

- Defined in externals.js as an ID → URL map.
- Sidebar router opens links in new tabs.
- Manifest only declares node type = 'link'.

11. Styling Philosophy

- Monospace.
- Black background.
- No hover highlights.
- Minimal affordances.
- Emoji used sparingly as semantic markers.
- Layout mirrors original static version exactly.

12. Intended Future Features (Not Yet Implemented)

- Pages controlling sidebar state (open folders, highlight nodes).
- Homepage as interactive 3D or generative environment.
- Babylon.js experiments.
- README pages for all projects.
- Mobile optimization pass.
- CV / About / Contact pages.

13. Development Philosophy

- Avoid frameworks unless absolutely necessary.
- Prefer explicit systems over abstraction layers.
- Accept known bugs if they unblock momentum.
- Optimize for clarity over completeness.
- Treat the site as a living system, not a finished artifact.

14. How to Help as a Coding Assistant

If you are an assistant reading this:

- Do not attempt to redesign architecture unless asked.
- Respect existing decisions even if suboptimal.
- Work incrementally.
- Assume the user prefers understanding over magic fixes.
- Avoid overengineering.

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